The impact of national health insurance on health services utilization in the Lao People's Democratic Republic (Lao PDR): a quasi-experimental evaluation using longitudinal administrative data

Appendix B. Online Supporting Information

2026-01-19

Table of contents

B.1. Scoping review of literature 1
B.2. Health financing landscape in Lao PDR 3
B.3. Financing arrangements before and after NHI 5
B.4. Data source, cleaning and imputation 8
B.5. Health service utilization trends 12
B.6. Robustness checks 15
B.7. Controlled interrupted time series analysis 17

List of tables

Table B1. Summary of health financing schemes in Lao PDR 4
Table B2. Summary of anticipated effects, mechanism and assumptions 5
Table B3. User fees at public health facilities in 2016 (in Lao kip) 6
Table B4. Provider payment arrangements for aged ≥5 before and after NHI 6
Table B5. Provider payment arrangements for aged <5 before and after NHI 6
Table B6. Provider payment arrangement for institutional births before and after NHI 7
Table B7. Health expenditure by financing scheme (million kip, inflation adjusted) 7
Table B8. Health inputs at the provincial level from 2014 to 2019 7
Table B9. Outcome variable details and data sources 8
Table B10. Covariate details and data sources 8
Table B11. Outliers and missing values in health service utilization data 10
Table B12. Imputation of outliers and missing values in health service utilization data 11
Table B13. Controlled interrupted time series analysis models 17

List of figures

Figure B1. Quarterly health service utilization trends 13
Figure B2. Quarterly health service utilization trends by intervention group 14
Figure B3. Sensitivity analysis on potential violation of parallel trends assumption 16

B.1. Scoping review of literature

In this scoping review, we looked for evidence on the impact of non-contributory health insurance schemes on health service utilization and financial risk protection among informal sector populations in low- and lower-middle-income countries (LLMICs). We searched PubMed on January 15, 2024 for studies published in English after 2009, using a combination of keywords and controlled vocabulary (Medical Subject Headings) related to four key topics: cost sharing, provider payment, health insurance, and impact. Using three pre-made methodological and geographic filters (Canadian Agency for Drugs and Technologies in Health; Canadian Agency for Drugs and Technologies in Health; Cochrane Effective Practice and Organisation of Care), the search was narrowed to focus on systematic reviews, randomized control trials, and quasi-experimental studies in low- and middle-income countries (LMICs) (World Bank, 2021). Studies focusing exclusively on upper-middle-income countries were manually excluded.

The search returned 5,724 records after removing duplicates. After screening the titles, abstracts and reference lists of selected records, 26 records were retained for a full-text review. However, none of the original studies, including those identified in three systematic reviews addressing similar research questions, met the final inclusion criteria.

A 2013 review identified 19 studies examining the impact of health insurance schemes for informal sector populations in LMICs, with eight conducted in four LLMICs (Acharya et al., 2012). A 2019 review, with a broader population focus, identified 68 studies (Erlangga et al., 2019). Of these, 33 were from LLMICs, and 19 specifically targeted or included non-poor informal sector populations. The most recent review, published in 2023 and focused on Southeast Asia, included two randomized controlled trials and four quasi-experimental evaluations from three LLMICs (Kaiser et al., 2023). However, all of the programs or schemes evaluated in these reviews were variations of social health insurance, voluntary health insurance, or subsidized schemes for vulnerable groups, rather than non-contributory models targeting non-poor populations in the informal sector.

Health financing literature in Lao PDR

Following our primary search, we also reviewed literature on health financing in Lao PDR. This search, covering initial PubMed results, article reference lists, and grey literature, yielded 29 records. The records comprised: (1) descriptive overviews of the health financing system (n = 2) (International Labour Organization, 2021; World Bank, 2017b); (2) analyses of trends in health service utilization and out-of-pocket spending using national survey data (2002/03−2018/19; n = 4) (National Institute of Public Health Lao PDR, 2017; Rannan-Eliya et al., 2012; World Bank, 2010c; World Health Organization Regional Office for the Western Pacific, 2023); and (3) studies focusing on specific programs, services, or themes (n = 23), including the Free Maternal, Newborn, and Child Health program (Ahissou et al., 2023; Boudreaux et al., 2014; Jeffery et al., 2015; Nagpal et al., 2019; World Bank, 2013), community-based health insurance (Alkenbrack et al., 2013; Alkenbrack and Lindelow, 2013; Bodhisane and Pongpanich, 2015, 2017, 2019a, 2019b; Sydavong et al., 2019; World Bank, 2010a), social health insurance (Alkenbrack et al., 2015; World Bank, 2010b), the National Health Insurance scheme (Bodhisane and Pongpanich, 2019b, 2022, 2023; Chaleunvong, Phoummalaysith, Phonvixay, Sychareun, et al., 2020; Chaleunvong, Phoummalaysith, Phonvixay, Vonglokham, et al., 2020; Ito et al., 2022), health system performance (Aryal et al., 2024a, 2024b) and health service delivery during the COVID-19 pandemic (Arsenault et al., 2022).

Of the six NHI-related studies, four examined the effects of NHI and found a higher likelihood of health service use, particularly among the chronically ill and elderly, while the findings on financial protection were mixed (Bodhisane and Pongpanich, 2019b, 2022, 2023; Ito et al., 2022). However, these studies were cross-sectional and conducted in only one province. The other two examined factors associated with payments exceeding the scheduled NHI fees and low levels of patient knowledge about the scheme (Chaleunvong, Phoummalaysith, Phonvixay, Sychareun, et al., 2020; Chaleunvong, Phoummalaysith, Phonvixay, Vonglokham, et al., 2020). At the national-level, a government-WHO report, based on three national socioeconomic surveys conducted between 2007/08 and 2018/19, found improvements in service use and financial protection over time but did not establish causal relationships (World Health Organization Regional Office for the Western Pacific, 2023).

More recent studies evaluated health system performance with a focus on quality and equity (Aryal et al., 2024a, 2024b), and examined how COVID-19 affected health service utilization (Arsenault et al., 2022). Although relevant, these studies had a broader scope than NHI and were conducted in the years following the pandemic. Overall, we found no rigorous evaluations of large-scale, non-contributory health insurance schemes targeting informal sector populations in LLMICs, including Lao PDR, on health and financial risk protection outcomes.

B.2. Health financing landscape in Lao PDR

After the communist revolution in 1975, health services were officially made freely available for all at public health facilities in Lao PDR (Akkhavong et al., 2014; National Institute of Public Health Lao PDR, 2005). However, the collapse of the centrally planned economy in the 1980s severely undermined the health system, causing shortages of funds, supplies, workforce and infrastructure (Akkhavong et al., 2014; Phommasack et al., 2005; Thomé and Pholsena, 2009). This left the government with little choice but to restructure and decentralize core health system functions to the sub-national level and adopt market-oriented approaches in its programs (Akkhavong et al., 2014).

In health financing, the free health care policy was abolished in 1995, formalizing the existing practice of user fees (Government of Lao PDR, 1995). Two years later, in 1997, the Revolving Drug Fund (RDF) was officially established and implemented in these facilities as a means of cost recovery for pharmaceuticals (Ministry of Health and Long-Term Care, 1997). Initially, these measures were intended to help health providers to cover the costs of their services from patients and replenish drugs through a mark-up on sales (Murakami et al., 2001; National Institute of Public Health Lao PDR, 2005). However, an inherent incentive for providers to maximize RDF revenues, coupled with persistently low government funding, ultimately led the health system to become overly reliant on out-of-pocket payments to finance health care.

In 2015, Laos' health system was primarily public in terms of service delivery but heavily financed by private and external sources (Akkhavong et al., 2014). Out-of-pocket (OOP) expenditures constituted the largest share of total health spending at 42.3%. Government and public health insurance spending combined accounted for 36.3%, while donors and non-government organizations contributed 21.3%. These proportions remained relatively stable through 2019, with OOP spending slowly decreasing to 39.9%, and government and health insurance spending increasing to 37.5% (Ministry of Health Lao PDR, 2021). The high level of OOP spending, approaching 40% of total health expenditure, remains one of the main drivers of household vulnerability and impoverishment in Lao PDR (World Bank, 2015, 2017a).

Prior to 2016, the health financing landscape in Lao PDR was complex, with multiple schemes covering different segments of the population (see Table B1) (Akkhavong et al., 2014). Public health insurance was formally introduced in 1995 through the Civil Servant Scheme (reformed as the State Authority for Social Security in 2006), which provided mandatory coverage for civil servants. Subsequently, in 2001, coverage was extended to employees of private companies through the voluntary Social Security Organization (SSO). Both schemes were financed by a 1.25% payroll tax, split evenly between employees and employers. By 2015, the two schemes together covered 7.5% of the population, mostly formal sector workers. Police and military personnel were covered under separate health insurance programs.

Table B1. Summary of health financing schemes in Lao PDR

Schemes (population and enrollment)	Financing (as of 2016)	Years of introduction
Contributory schemes†		
SASS: State Authority for Social Security (civil servants and their dependants; mandatory enrollment) − formerly Civil Servant Scheme (CSS)	1.25% monthly payroll tax (0.625% employee + 0.625% employer)	1995 (formalized) 2006 (reformed)
SSO: Social Security Organization (private sector workers, the self-employed, and their dependents; voluntary enrollment at the employer level, individual enrollment allowed from 2015)	1.25% monthly payroll tax (0.625% employee + 0.625% employer)	2001
CBHI: Voluntary community-based health insurance schemes (informal sector population, including self-employed, informal wage earners, and their family members; voluntary enrollment at the household level)	Monthly premium based on family size (300,000 kip for 1 person to 700,000 kip >8 persons household)	2002 (pilot) 2006 (extension)
Non-contributory schemes		
HEF: Health Equity Fund (poor households; eligibility based on the national poverty line or other mean-testing approaches)	Donor assistance and government revenues	2004
Free MNCH: Free Maternal, Newborn, and Child Health program (pregnant women and children under five)	Donor assistance and government revenues	2012 (official)
NHI: National Health Insurance (all informal sector population without public health insurance, excluding residents of Vientiane Capital; universal entitlement)	Government revenues with co-payments	2016

Note: † Data for police and military schemes are excluded. Source: Progress towards Universal Health Coverage in the Lao PDR: Monitoring financial protection 2007−2019 (World Health Organization Regional Office for the Western Pacific, 2023).

Meanwhile, health insurance for the large informal sector population expanded slowly and ineffectively through a patchwork of schemes (Akkhavong et al., 2014; Thomé and Pholsena, 2009). Voluntary community-based health insurance (CBHI) was first piloted in 2002 and gradually expanded in 2006. By 2015, it operated in 50 districts (35% of total districts), targeting individuals in the informal economy, including informal wage workers, the self-employed, and their dependents. However, population coverage remained low at just 2.2%, with problems of low enrollment, adverse selection, inadequate risk pooling, and poor management. The expansion of SSO in 2015 to include informal sector workers also did not meaningfully increase coverage, reaching less than 1% of the population (International Labour Organization, 2024).

To address coverage gaps, two targeted schemes were introduced. The Health Equity Fund (HEF) was launched in 2005, with donor support, aiming to provide free health care and social benefits to households below the national poverty line (US$0.85 per person per day). Around the same time, free packages of services for pregnant women and children under the age of five (CU5) were also provided sporadically by donors in various regions, either in conjunction with HEF or separately, before being scaled up in 2012 as the Free Maternal, Newborn and Child Health (Free MNCH) Program with official government support. By 2015, both schemes had been gradually expanded and implemented in 125 and 142 districts, respectively.

Despite these scale-ups, the overall coverage and effectiveness of these schemes remained limited. In 2013, 24.6% of the population lived below the national poverty line; yet, by 2015, the HEF reached only 6.1%. Similarly, even with free services, only about half of live births were estimated to be facility-based in 2015 (Ministry of Health Lao PDR, 2016), with considerable disparities in uptake and financial protection across socioeconomic groups (Ahissou et al., 2023; Nagpal et al., 2019). In 2015, these three schemes combined (CBHI, HEF, Free MNCH) covered just 16.9% of the population. When formal sector schemes were included, national population coverage reached approximately 33%.

B.3. Financing arrangements before and after NHI

Impact of NHI on health service utilization

In this study, we examined whether simultaneous changes in provider payment mechanisms and demand-side arrangements under NHI affected health service utilization. Table B2 outlines the anticipated effects, underlying mechanisms, and key assumptions for each study outcome.

Table B2. Summary of anticipated effects, mechanism and assumptions

Outcome	Anticipated effect	Mechanism	Assumption
Outpatient services aged ≥5	Increase in utilization	Demand-side: reduction in user fees Supply-side: shift from user fees to capitation at both hospitals and health centers	1) The existing CBHI and HEF schemes had little to no impact on service utilization; 2) No spillover effects of NHI on formal sector health insurance scheme members
Inpatient services aged ≥5	Increase in utilization	Demand-side: reduction in user fees Supply-side: shift from user fees to case-based payments at hospitals and capitation at health centers	
Outpatient services aged <5	Increase in utilization	Demand-side: no change Supply-side: introduction of capitation; possible improvement in compliance with the exemption mechanism, which may reduce user fees	The Free MNCH program did not subsidize outpatient care for children (aged <5), which likely contributed to under-provision of services and weak enforcement of the exemption mechanism
Inpatient services aged <5	Increase in utilization	Demand-side: no change Supply-side: no change, except for the introduction of capitation at health centers (overlapping with Free MNCH case-based funding); limited case-based payment for services not covered by the Free MNCH program; possible improvement in compliance with the exemption mechanism, which may reduce user fees	Inpatient care for children (aged <5) were subsidized under the Free MNCH program, but there may have been under-provision of services due to erratic and/or inadequate funding and weak enforcement of the exemption mechanism
Inpatient length of stay	Decrease in average length of stay	Perceived inadequacy of NHI payment compared to user fees; pressure to contain costs in response to increased service use	
Institutional delivery	Increase in utilization	Demand-side: no change Supply-side: no change, except for the introduction of capitation at health centers (overlapping with Free MNCH case-based funding); limited case-based payment for services not covered by the Free MNCH program; possible improvement in compliance with the exemption mechanism, which may reduce user fees	Institutional deliveries were fully subsidized under the Free MNCH program, but there may have been under-provision of services due to erratic and/or inadequate funding and weak enforcement of the exemption mechanism

Demand-side financing arrangement

Table B3 details patient charges for outpatient and inpatient services at each facility level before and after NHI. The pre-NHI fees indicate the average expected patient charges per service before NHI, while the post-NHI fees represent the fixed co-payments introduced under NHI. These fees remained unchanged throughout the study period. Assuming good policy compliance by health providers, NHI is estimated to have reduced patient charges by 56%−72% for outpatient visits and 69%−93% for inpatient admissions at the point of service.

Free maternal and child health services were expanded in 2012, but subsidies remained inconsistent and insufficient in some places and did not cover outpatient services. This may have weakened the exemption mechanism, with some pregnant women and children ending up paying for services that were supposed to be free. In this context, NHI funding, particularly capitation, may have helped strengthen service delivery, enforce the exemption mechanism, and reduce informal payments. See Tables B5 and B6 for a detailed breakdown of provider payment arrangements for this target group.

Table B3. User fees at public health facilities in 2016 (in Lao kip)

	User fees per outpatient visit			User fees per inpatient admission			Targeted population§
Facility level	pre-NHI†	post-NHI‡	% change	pre-NHI	post-NHI	% change	
Provincial hospital	53,685	15,000	-72%	423,823	30,000	-93%	Free
District hospital	22,643	10,000	-56%	96,053	30,000	-69%	Free
Health center	12,074	5,000	-59%	36,448	5,000	-86%	Free

Note: The average exchange rate for 2016 was 1 Lao kip = 0.00012 US dollars, or conversely, 1 US dollar = 8,124 Lao kip. † Pre-NHI user fee estimates are based on a top-down costing of selected public health facilities in 2012/13 (Lao Health Costing Commitee, 2014). These fees represent the net cost of health services, excluding government-covered expenses such as salaries of medical professionals. The fees are adjusted for inflation to 2016 values using the health care component of the Consumer Price Index (Turner et al., 2019). ‡ Post-NHI fees represent the fixed patient co-payment set under NHI (Ministry of Health Lao PDR, 2018). § The target population includes pregnant women, children under five, and poor households living below the national poverty line (US$ 0.85 per person per day in 2012/13). Members of formal sector schemes are also exempt from co-payments.

Supply-side financing arrangement

Table B4 shows provider payment arrangements for individuals aged ≥5 by service type and facility level. Before NHI, health care providers operated on a fee-for-service model, requiring direct payment for all services. After NHI, outpatient services shifted to capitation (a fixed per-person payment for individuals in the facility's catchment area). Capitation rates are adjusted annually based on the previous year's utilization, with higher amounts allocated to facilities that served more patients. Hospitals also receive additional payments for minor surgeries and referred cases, though in limited numbers. For inpatient admissions, NHI implemented a case-based reimbursement model (a fixed payment per admission), except for health centers, which are fully capitated.

Table B4. Provider payment arrangements for aged ≥5 before and after NHI

	Outpatient visit		Inpatient admission	
Facility level	pre-NHI	post-NHI†	pre-NHI	post-NHI
Provincial hospital	No subsidies	Capitation	No subsidies	Case-based payment
District hospital	No subsidies	Capitation	No subsidies	Case-based payment
Health center	No subsidies	Capitation	No subsidies	Capitation

Note: † Capitation amounts vary, ranging from 4,000 to 25,000 kip for health centers, 4,000 to 37,000 kip for district hospitals, and 9,000 to 80,000 kip for provincial and regional hospitals. Hospitals also receive additional payments for certain services, including minor surgeries and referrals. These cases represented approximately 1% of total NHI services volume and 4% of total NHI claim amount.

Table B5 shows provider payment arrangements for children aged <5 before and after NHI. Since 2012, the Free MNCH program fully subsidized inpatient services for young children through case-based payments, but did not subsidize outpatient services, except in a handful of donor-supported districts. The program also provided small cash benefits to cover transportation and minor indirect expenses. After NHI, hospital outpatient services and all services at health centers were fully subsidized through capitation. Meanwhile, hospital inpatient services transitioned to NHI funding in October 2017, continuing case-based reimbursement but at increased rates. This change, however, occurred outside the period of our analysis (October 2015 − September 2017) and therefore should not have theoretically influenced the utilization of services for children aged <5.

Table B5. Provider payment arrangements for aged <5 before and after NHI

	Outpatient visit		Inpatient admission	
Facility level	pre-NHI	post-NHI (at rollout)	pre-NHI	post-NHI (Oct/2017)
Provincial hospital	No subsidies	Capitation	Case-based payment	Case-based payment
District hospital	No subsidies	Capitation	Case-based payment	Case-based payment
Health center	No subsidies	Capitation	Case-based payment (or capitation)	Capitation

Note: The free health program for children under the age of five was implemented in 138 districts by Q4 2015.

Table B6 shows provider payment arrangements for institutional births before and after NHI. Case-based payments continued in both periods, but increased after NHI, except for health centers, which switched to capitation. As with services for children under five, however, this change occurred in October 2017. Thus, for the period we analyzed (October 2015 − September 2017), we do not expect a significant impact of NHI payment reform on institutional birth.

Table B6. Provider payment arrangement for institutional births before and after NHI

	Institutional birth	
Facility level	pre-NHI†	post-NHI (Oct/2017)
Provincial hospital	Case-based payment	Case-based payment
District hospital	Case-based payment	Case-based payment
Health center	Case-based payment	Capitation

Note: † Provider payments under the Free MCH scheme varied across regions due to different funding schemes implemented by donors.

Table B7 presents trends in heath expenditure by financing scheme from 2014 to 2019, adjusted for inflation to 2019 values. Between 2015 and 2018, public health financing scheme expenditure (social health insurance, community-based health insurance, national health insurance and others) increased from 59 billion to 318 billion kip, with NHI contributing 88% of this growth. In 2018, NHI accounted for 72% of SHP expenditure or 6% of current health expenditure.

Table B7. Health expenditure by financing scheme (million kip, inflation adjusted)

	2014	2015	2016	2017	2018	2019
Central government	656,187	920,859	983,125	729,523	440,060	649,171
Provincial government	524,078	615,421	551,933	864,941	978,659	1,200,154
National health insurance	-	-	-	127,416	228,663	479,840
Social health insurance	47,094	53,577	76,915	89,341	86,587	44,464
Community-based health insurance	6,670	5,595	4,068	-	3,223	1,072
Non-governmental organization	84,759	102,162	121,722	203,405	82,911	94,424
Out-of-pocket expenditure	1,366,011	1,409,595	1,506,882	1,732,617	1,717,231	1,775,857
Total	2,684,799	3,107,209	3,244,644	3,747,243	3,537,333	4,244,982

Note: Figures are adjusted for inflation to 2019 values; government expenditure includes donor contributions that are channeled through the government system, which totaled 806 billion kip in 2019. Source: Lao National Health Accounts 2019 (Ministry of Health Lao PDR, 2021).

Supply-side resources

At the provincial level, supply-side health inputs changed little before and after the introduction of NHI (Table B8) (Ministry of Health Lao PDR, 2022). The number of beds per 1000 remained stable between 2015 and 2019 at about 0.61 for hospitals and 0.43 for health centers. The number of physicians per 1000 increased slightly from 0.45 to 0.53 between 2016 and 2019, while the number of nurses per 1000 decreased from 1.05 to 0.95. The number of midwives remained steady, averaging 0.26 per 1000 over the same period.

Table B8. Health inputs at the provincial level from 2014 to 2019

	2014	2015	2016	2017	2018	2019
Public hospital beds per 1000	na	0.62	0.61	0.60	0.60	0.61
Health center beds per 1000	na	0.43	0.43	0.42	0.43	0.44
Physicians per 1000	na	na	0.45	0.53	0.54	0.53
Nurses per 1000	na	na	1.05	0.97	0.97	0.95
Midwives per 1000	na	na	0.25	0.27	0.26	0.25

Note: na = not available. Source: National Health Statistics Report 2020 (Ministry of Health Lao PDR, 2022) and District Health Information System 2.

B.4. Data source, cleaning and imputation

Data sources

Data for this study was drawn from the Lao District Health Information System 2 (DHIS2), a routine health information system of the government that stores and provides access to a variety of health data, including facility-level aggregated statistics, collected and reported by all public health facilities in Lao PDR. Additional data, including district-level population estimates and socioeconomic information, were obtained from government reports (Coulombe et al., 2016; Lao Statistics Bureau, 2015). Table B9 and Table B10 detail the definitions, sources, and other specifics of the outcome variables and covariates used in this study.

The main analyses used health service utilization data from district hospitals and health centers in 17 provinces spanning 24 months from October 2015 to September 2017. Sensitivity analyses used a broader dataset, including provincial hospitals and Vientiane Capital. Of the 139 districts that implemented NHI, 129 were included in the final analyses. The excluded group comprised nine districts that piloted variations of NHI from 2015 to 2016 (Met, Viangkham, Mun, Nong, Samouay, Lamam, Kalum, Dakchung, and Thateng) and one district (Phongsali) that did not implement the Free MNCH program by the fourth quarter of 2015.

We noted that the outpatient visits variable excluded follow-up visits, treatment provided during outreach activities, health check-ups, and visits to obtain a health certificate. We further deducted an estimated four rounds of antenatal care visits and two rounds of postnatal care visits to avoid possible double counting. Institutional births were also excluded from the number of inpatient admissions. These deductions did not significantly alter our final DiD estimates. The inpatient admission variable was treated as a single discharge episode, given the absence of a specific follow-up variable in DHIS2.

Table B9. Outcome variable details and data sources

Variable	Nominator (utilization)	Denominator (population)	Stratification	Frequency / Level	Source / Date of access
Outpatient visit rate	Number of outpatient visits	Population estimates	Age (<5, ≥5)	Monthly facility-level data were aggregated to quarterly district-level data	Utilization data were downloaded from DHIS2 in December 2023. Population estimates, originally from the Lao Statistics Bureau's 2015 Population Census (Lao Statistics Bureau, 2015), were also downloaded from DHIS2.
Inpatient admission rate	Number of inpatient admissions	Population estimates	Age (<5, ≥5)		
Average length of days in inpatient beds	Number of inpatient days	Number of Inpatient admissions	Not available		
Institutional birth rate	Number of births at facility	Estimated number of live births	Not relevant		

DHIS2 = District Health Information System 2.

Table B10. Covariate details and data sources

Variable	Frequency / Level	Source / Date of access
% of population who are self-employed	District-level data	2015 Lao Census-based Poverty Map (Coulombe et al., 2016)
% of population living in urban area		
% of population with access to improved sanitation		
% of population aged 15-64 who are literate		
Poverty headcount ratio		
Number of health facility	Monthly facility-level data were aggregated to quarterly district-level data	Based on the number of facilities reporting outpatient cases in DHIS2

DHIS2 = District Health Information System 2.

Data quality

In our preparatory analysis, we examined DHIS2 utilization trends starting in Q4 2013, when DHIS2 rollout began (Chu et al., 2017), to determine the appropriate starting point for the analysis. We observed gradual improvements in the completeness of outpatient and inpatient reporting over time and identified the period beginning in Q4 2015 as providing sufficiently reliable data.

For data quality, we assessed the completeness and internal consistency of monthly facility-level data from October 2015 to March 2019 for each health facility included in the analyses (MEASURE Evaluation, 2020; Wagenaar et al., 2016). External consistency of data was not verified due to the lack of reliable data sources for comparison.

Data completeness was evaluated by verifying the presence of all required data points within the dataset. Between October 2015 and March 2019, hospitals demonstrated an overall completeness rate of 97.4% in the pre-NHI period and 99.2% in the post-NHI period (Table B11). In constrast, health centers showed a relatively lower completeness rate of 80.5% and 81.7% in the pre- and post-NHI periods, respectively. Given that zeros constitute less than 1% of all data, the missing values likely represent unrecorded zeros rather than true absence of data. This issue was accounted for in our subsequent imputation strategy.

Internal consistency was assessed by identifying values that were extreme relative to other values in the series. We first divided the data into pre- and post-NHI periods and used Friedman's super smoother algorithm to remove seasonal effects and underlying trends (Friedman, 1984). Potential outliers were then identified using Tukey's 'far out' method (Tukey, 1977), which defines outliers as observations exceeding three times the interquartile range (IQR) below the first quartile (Q1) or above the third quartile (Q3). The data were also screened for nonsensical cases by examining the coherence between related indicators. These included instances such as outpatient visits for children aged <5 exceeding those for aged ≥5, or inpatient admissions exceeding outpatient visits. In total, 3.2% of the data were identified as potential outliers (Table B11).

Table B11. Outliers and missing values in health service utilization data

	Pre-NHI			Post-NHI		
	Total	Outlier detected	Missing values	Total	Outlier detected	Missing values
	Hospital					
Antenatal visit 1	3099 (100%)	56 (1.8%)	25 (0.8%)	3220 (100%)	42 (1.3%)	5 (0.2%)
Antenatal visit 4	3099 (100%)	55 (1.8%)	45 (1.5%)	3220 (100%)	42 (1.3%)	11 (0.3%)
Institutional delivery	3099 (100%)	57 (1.8%)	121 (3.9%)	3220 (100%)	40 (1.2%)	4 (0.1%)
Postnatal visit	3099 (100%)	54 (1.7%)	219 (7.1%)	3220 (100%)	50 (1.6%)	68 (2.1%)
Outpatient aged ≥5	3099 (100%)	79 (2.5%)	2 (0.1%)	3220 (100%)	61 (1.9%)	0 (0.0%)
Outpatient aged <5	3099 (100%)	76 (2.5%)	3 (0.1%)	3220 (100%)	68 (2.1%)	1 (0.0%)
Inpatient aged ≥5	3099 (100%)	121 (3.9%)	51 (1.6%)	3220 (100%)	91 (2.8%)	12 (0.4%)
Inpatient aged <5	3099 (100%)	51 (1.6%)	146 (4.7%)	3220 (100%)	69 (2.1%)	55 (1.7%)
Inpatient day	3099 (100%)	80 (2.6%)	129 (4.2%)	3220 (100%)	49 (1.5%)	67 (2.1%)
Sub-total	27891 (100%)	629 (2.3%)	741 (2.7%)	28980 (100%)	512 (1.8%)	223 (0.8%)
	Health center					
Antenatal visit 1	20210 (100%)	518 (2.6%)	2044 (10.1%)	22514 (100%)	470 (2.1%)	1948 (8.7%)
Antenatal visit 4	20210 (100%)	592 (2.9%)	3957 (19.6%)	22514 (100%)	711 (3.2%)	4116 (18.3%)
Institutional delivery	20210 (100%)	807 (4.0%)	7933 (39.3%)	22514 (100%)	853 (3.8%)	7490 (33.3%)
Postnatal visit	20210 (100%)	825 (4.1%)	7216 (35.7%)	22514 (100%)	795 (3.5%)	6290 (27.9%)
Outpatient aged ≥5	20210 (100%)	404 (2.0%)	149 (0.7%)	22514 (100%)	516 (2.3%)	57 (0.3%)
Outpatient aged <5	20210 (100%)	442 (2.2%)	454 (2.2%)	22514 (100%)	446 (2.0%)	170 (0.8%)
Inpatient aged ≥5	20210 (100%)	585 (2.9%)	3586 (17.7%)	22514 (100%)	561 (2.5%)	2438 (10.8%)
Inpatient aged <5	20210 (100%)	1316 (6.5%)	10470 (51.8%)	22514 (100%)	1897 (8.4%)	12292 (54.6%)
Inpatient day	20210 (100%)	571 (2.8%)	4585 (22.7%)	22514 (100%)	667 (3.0%)	3357 (14.9%)
Sub-total	181890 (100%)	6060 (3.3%)	40394 (22.2%)	202626 (100%)	6916 (3.8%)	38158 (21.0%)
Total	209781 (100%)	6689 (3.2%)	41135 (19.6%)	231606 (100%)	7428 (3.5%)	38381 (18.3%)
Total (pre- and post-NHI combined)					14117 (3.2%)	79516 (18.0%)

Outliers and missing values were replaced with linearly interpolated values estimated from deseasonalized and detrended neighboring observations. We treated outliers and missing values differently for hospital and health center data. For hospitals, all identified outliers and missing values were replaced. In constrast, for health centers, outliers were replaced only if the series had fewer than four missing values and if the median exceeded pre-defined quartile thresholds (0%, 25%, 50%, 75%) of all health centers. This stricter approach minimized the risk of imputing missing values that could potentially be zero, especially in health centers with low service utilization. In total, 34.7% of potential outliers and 2.7% of missing data were replaced, representing 1.1% and 0.5% of the total data, respectively (Table B12).

Table B12. Imputation of outliers and missing values in health service utilization data

	Pre-NHI			Post-NHI		
	Total	Outlier replaced	Missing replaced	Total	Outlier replaced	Missing replaced
	Hospital					
Antenatal visit 1	3099 (100%)	56 (1.8%)	25 (0.8%)	3220 (100%)	42 (1.3%)	5 (0.2%)
Antenatal visit 4	3099 (100%)	55 (1.8%)	45 (1.5%)	3220 (100%)	42 (1.3%)	11 (0.3%)
Institutional delivery	3099 (100%)	57 (1.8%)	121 (3.9%)	3220 (100%)	40 (1.2%)	4 (0.1%)
Postnatal visit	3099 (100%)	54 (1.7%)	219 (7.1%)	3220 (100%)	50 (1.6%)	68 (2.1%)
Outpatient aged ≥5	3099 (100%)	79 (2.5%)	2 (0.1%)	3220 (100%)	61 (1.9%)	0 (0.0%)
Outpatient aged <5	3099 (100%)	76 (2.5%)	3 (0.1%)	3220 (100%)	68 (2.1%)	1 (0.0%)
Inpatient aged ≥5	3099 (100%)	121 (3.9%)	9 (0.3%)	3220 (100%)	91 (2.8%)	12 (0.4%)
Inpatient aged <5	3099 (100%)	51 (1.6%)	104 (3.4%)	3220 (100%)	69 (2.1%)	55 (1.7%)
Inpatient day	3099 (100%)	80 (2.6%)	76 (2.5%)	3220 (100%)	49 (1.5%)	62 (1.9%)
Sub-total	27891 (100%)	629 (2.3%)	604 (2.2%)	28980 (100%)	512 (1.8%)	218 (0.8%)
	Health center					
Antenatal visit 1	20210 (100%)	76 (0.4%)	29 (0.1%)	22514 (100%)	59 (0.3%)	24 (0.1%)
Antenatal visit 4	20210 (100%)	69 (0.3%)	47 (0.2%)	22514 (100%)	59 (0.3%)	30 (0.1%)
Institutional delivery	20210 (100%)	68 (0.3%)	59 (0.3%)	22514 (100%)	49 (0.2%)	42 (0.2%)
Postnatal visit	20210 (100%)	65 (0.3%)	49 (0.2%)	22514 (100%)	19 (0.1%)	38 (0.2%)
Outpatient aged ≥5	20210 (100%)	401 (2.0%)	60 (0.3%)	22514 (100%)	515 (2.3%)	30 (0.1%)
Outpatient aged <5	20210 (100%)	279 (1.4%)	90 (0.4%)	22514 (100%)	328 (1.5%)	27 (0.1%)
Inpatient aged ≥5	20210 (100%)	354 (1.8%)	224 (1.1%)	22514 (100%)	359 (1.6%)	194 (0.9%)
Inpatient aged <5	20210 (100%)	298 (1.5%)	17 (0.1%)	22514 (100%)	334 (1.5%)	37 (0.2%)
Inpatient day	20210 (100%)	180 (0.9%)	102 (0.5%)	22514 (100%)	239 (1.1%)	187 (0.8%)
Sub-total	181890 (100%)	1790 (1.0%)	677 (0.4%)	202626 (100%)	1961 (1.1%)	609 (0.3%)
Total	209781 (100%)	2419 (1.2%)	1281 (0.6%)	231606 (100%)	2473 (1.2%)	827 (0.4%)
Total (pre- and post-NHI combined)					4892 (1.1%)	2108 (0.5%)

We investigated how different approaches to handling outliers and missing data could affect our DiD estimates. Two models were considered: (a) no imputation of missing values and outliers, and (b) imputation of all missing values as zero. The model without imputation generally produced slightly higher DiD estimates with wider confidence intervals, suggesting that our main model offers more conservative and precise estimates of NHI effects. In comparison, the zero-imputation model produced estimates almost identical to those of the main imputation model.

B.5. Health service utilization trends

Figure B1 presents quarterly trends in health service utilization outcomes, spanning seven quarters before ($e = -7$) and five quarters after ($e = 4$) the introduction of NHI ($e = 0$). During the pre-NHI period, utilization trends were largely parallel between the intervention and control groups for all six outcomes. After NHI, outpatient visits (aged <5, ≥5) and inpatient admissions (aged ≥5) increased significantly in the intervention group compared with the control group, relative to the baseline quarter ($e = -1$).

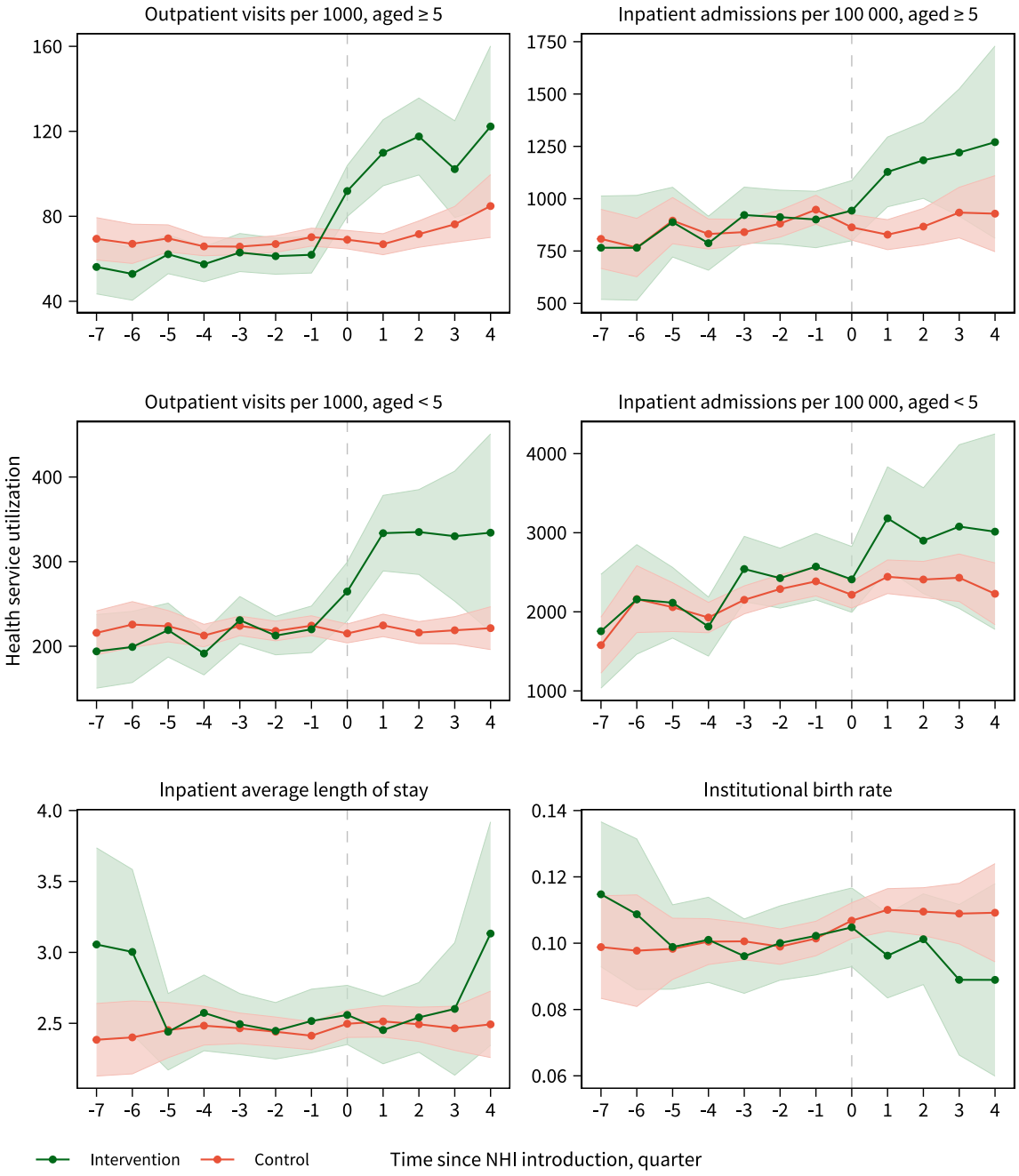
Outpatient visits per 1000 individuals diverged immediately at NHI rollout. Among individuals aged ≥5, visits increased in the intervention group (62 → 92, +49%) but decreased slightly in the control group (70 → 69, −2%). A similar pattern was observed for children aged <5, with visits increasing in the intervention group (220 → 265, +20%) while decreasing in the control group (224 → 215, −4%).

Inpatient admissions per 100 000 individuals diverged two quarters after NHI implementation ($e = 1$). Among individuals aged ≥5, admissions increased in the intervention group (901 → 1128, +25%) but decreased in the control group (947 → 828, −13%). Admissions for children aged <5 increased in both groups, but more significantly in the intervention group (2572 → 3182, +24%) than in the control group (2385 → 2443, +2%).

The average inpatient length of stay followed parallel trends before NHI and changed minimally after implementation, increasing by 2% (2.5 to 2.6) in the intervention group and by 3% (2.4 to 2.5) in the control group. Institutional birth rates decreased over five quarters ($e = 0$–4) in the intervention group (0.10 to 0.09, −12%) but increased slightly in the control group (0.10 to 0.11, +8%). The differential trends at $e = -7$, −6 and 4 likely reflect existing baseline differences in utilization levels across intervention periods, as further illustrated in Figure B2.

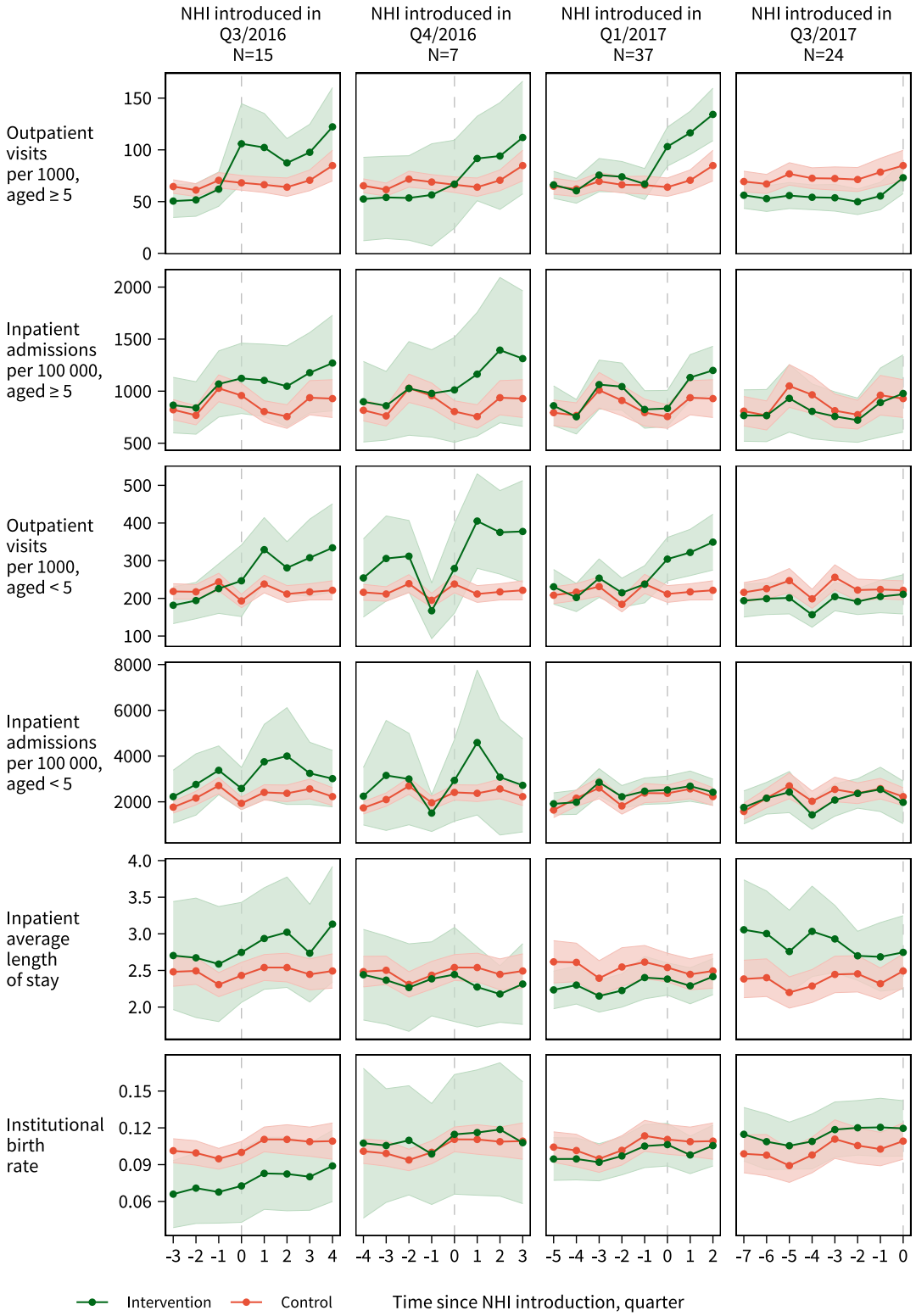
Figure B2 presents the utilization trends disaggregated by intervention group (i.e., implemention phase). Before NHI, trends were largely parallel across all six outcomes in all four phases, except for a decline in inpatient admissions in Q4 2016. Consistent with Figure B1, utilization increased substantially in the intervention group during the first three phases of NHI expansion, particularly for outpatient visits (aged <5, ≥5) and inpatient admissions among individuals aged ≥5. Inpatient admissions for children aged <5 also increased during the first two phases. In contrast, trends in average inpatient length of stay and institutional birth rates remained relatively stable across all phases.

Figure B1. Quarterly health service utilization trends



Note: The circles represent the average utilization of the outcome variable, and the shaded areas indicate the 95% pointwise confidence intervals. These are plotted against time relative to NHI introduction on the x-axis. The gray vertical dashed line at event time $e = 0$ marks the introduction of NHI. Source: Lao District Health Information System 2 and 2015 Lao Population Census.

Figure B2. Quarterly health service utilization trends by intervention group



B.6. Robustness checks

Sensitivity to violations of parallel trends assumption

Using the method of Rambachan and Roth (2023), we tested the sensitivity of our DiD estimates to potential violations of the parallel trends assumption. Rather than assuming perfect parallel trends in outcomes, this method allows for some deviations in parallel trends by imposing restrictions on how much the post-treatment violation can exceed that observed in the pre-treatment period. Our sensitivity analyses focused on three key outcomes: outpatient visits for individuals aged ≥5 and children aged <5, and inpatient admissions for individuals aged ≥5. Figure B3 shows robust 95% confidence intervals for the DiD coefficients estimated under varying relative magnitude ($\bar{M}$) and smoothness (M) restrictions.

(i) Relative magnitude restrictions

The left column of Figure B3 displays robust confidence intervals across varying values of $\bar{M}$, which is a multiplier for the relative magnitude (RM) restriction. This restriction formalizes the idea that the magnitude of post-treatment violations of parallel trends is bounded by $\bar{M}$ times the maximum violation observed in the pre-treatment period. We set $\bar{M}$ = 1, given our DiD models already accounted for observed covariates and used 'later-treated' districts as the control group. Under these conditions, we reasoned that the magnitude of any remaining unobserved differential trend in the post-NHI period would not likely exceed that in the pre-NHI period.

For outpatient visits among individuals aged ≥5, the 95% robust confidence intervals excluded zero at $\bar{M} = 2$, indicating that the post-NHI estimates remained statistically significant under violations up to twice the magnitude of the pre-NHI parallel trend violation. For children aged <5, the confidence interval excluded zero at $\bar{M} = 1$, indicating that the results remained robust to post-NHI violations as large as the maximum violation observed in the pre-NHI period. In contrast, for inpatient admissions among individuals aged ≥5, the confidence interval excluded zero only when $\bar{M} = 0$, suggesting that the results were valid only under the strict assumption of conditional parallel trends. Even a small violation could invalidate the statistical significance of the results.

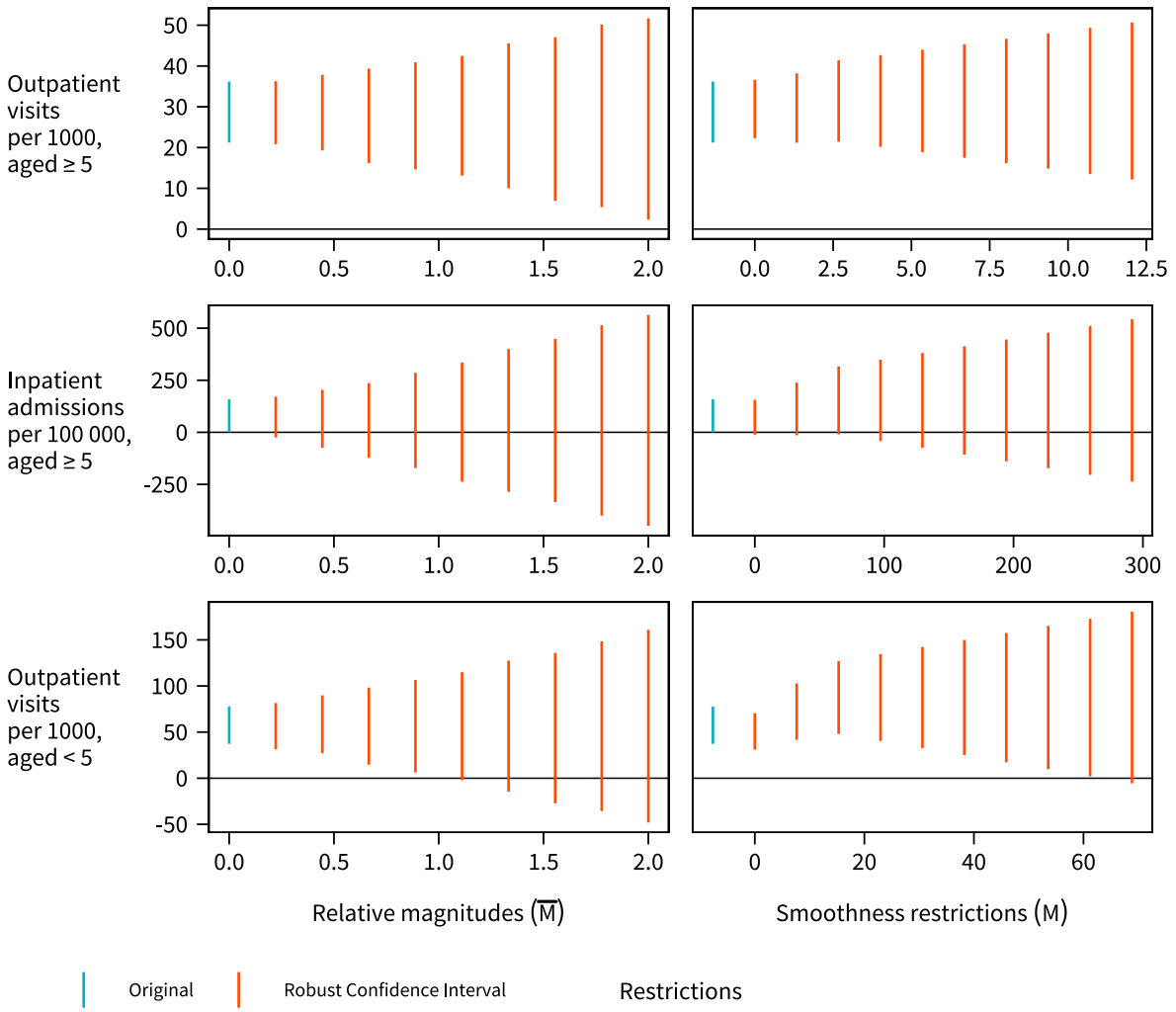
(ii) Smoothness restrictions

The right column of Figure B3 presents robust confidence intervals across varying values of M, which is the absolute bound imposed under the smoothness (SD) restriction. This restriction formalizes the idea that the slope of the pre-treatment differential trend cannot deviate nonlinearly by more than M. We set M = 0 for the same reasons discussed above, and based on visual and parametric assessments of the pre-NHI trend. We reasoned that any remaining unobserved confounding that evolved smoothly over time would not likely introduce significant nonlinearity into the post-NHI trends.

For outpatient visits among individuals aged ≥5, our DiD estimates remained robust with deviations greater than 12.5, which is about 28% of the DiD estimate of 44. For children aged <5, deviations up to 60, or about 53% of the DiD estimate of 112, still yielded robust confidence intervals that exclude zero. In contrast, for inpatient admissions among individuals aged ≥5, statistical significance was maintained only under strict linearity (M= 0), suggesting that any nonlinear deviation in the post-NHI trend would likely invalidate our results.

Overall, our causal interpretations of the DiD estimates for outpatient visits (aged <5 and ≥5) remained robust to plausible violations of the parallel trends assumption, even under moderate magnitudes of violation and non-linearity in trends. In constrast, estimates for inpatient admissions (aged ≥5) appeared sensitive to such deviations and should be interpreted with caution.

Figure B3. Sensitivity analysis on potential violation of parallel trends assumption



Note: The blue vertical lines represent original 95% confidence intervals for the difference-in-differences (DiD) estimates, while the red vertical lines represent robust 95% confidence intervals, estimated under varying relative magnitude ($\overline{M}$) and smoothness (M) restrictions. The breakdown point is the smallest restriction value where the confidence interval includes zero. DiD estimates are no longer considered statistically significant beyond this point. Source: Lao District Health Information System 2 and 2015 Lao Population Census for outcome data; 2015 Lao Poverty Map Report for covariates.

B.7. Controlled interrupted time series analysis

As a robustness check, we employed a multi-level controlled interrupted time series (CITS) design with a segmented regression to estimate the effect of NHI (Kontopantelis et al., 2015; Lopez Bernal et al., 2018). Four models were specified using combinations of monthly (16 months) and quarterly (6 quarters) data, with two control groups: districts that introduced NHI in the fourth quarter of 2017 (last-treated) and districts in Vientiane Capital that never introduced NHI (never-treated). All models used robust standard errors that account for potential heteroskedasticity and autocorrelation in the error terms (Arellano, 1987; Bottomley et al., 2019). Full details of the model specification are provided in Table B13.

For each model, we estimated Equation 1 for district d at time t, separately for each province:

$$Y_{dt} = \beta_0 + \beta_1 Time_t + \beta_2 Post_t + \beta_3 Time_t * Post_t + \beta_4 Intv_d + \beta_5 Invt_d * Time_t + \beta_6 Intv_d * Post_t + \beta_7 Intv_d * Time_t * Post_t + \beta_8 X_{dt} + v_d + \varepsilon_{dt} \quad (1)$$

Here, Y_{dt} represents the utilization outcome of interest, $Time_t$ denotes the event time, $Post_t$ is a dummy for the post-NHI period, and $Intv_d$ is a dummy for the treated districts. X_{dt} represents the number of health facilities in the district and seasonal dummies, v_d denotes the district-specific random effect, and ε_{dt} is the independently distributed error term.

In this specification, β_5 tests the parallel trends assumption by examining differential trends in the pre-NHI period, while β_6 and β_7 capture the immediate level and slope differences, respectively, in utilization levels between the intervention and control groups in the post-NHI period. Each coefficient was then pooled separately across provinces using meta-analysis to obtain aggregated estimates.

The CITS estimates, shown in Figure B11, broadly corroborate our DiD estimates. Model 1, which used last-treated districts as controls, showed level and slope changes that closely mirrored our main DiD estimates. This alignment was consistent across all outcomes for both types of data. Model 2, employing never-treated districts as controls, also generally supported these findings.

Table B13. Controlled interrupted time series analysis models

	Model 1-1	Model 1-2	Model 2-1	Model 2-2
Control group	Last treated districts (46 districts in 5 provinces treated in October 2017)		Untreated districts† (3-4 selected districts in the capital city of Vientiane)	
Intervention Group	Early treated districts (59 districts in 9 provinces treated by February 2017)		All treated districts (129 districts in 16 provinces treated by October 2017)	
Data Frequency	Quarterly	Monthly	Quarterly	Monthly
Analysis period	6 quarters (3 quarters before and after NHI)	16 months (8 months before and after NHI)	6 quarters (3 quarters before and after NHI)	16 months (8 months before and after NHI)

Note: † Free maternal and child health services were introduced in some districts of Vientiane Capital as early as 2012, and in all remaining districts by January 2017. For this reason, Model 2 results for outpatient and inpatient care for children under five, average inpatient length of stay, and institutional births are not reported.

References

Acharya, A., Vellakkal, S., Taylor, F., Masset, E., Satija, A., Burke, M., et al. (2012). The Impact of Health Insurance Schemes for the Informal Sector in Low- and Middle-Income Countries: A Systematic Review. *The World Bank Research Observer, 28*(2), 236–266. doi:10.1093/wbro/lks009

Ahissou, N. C. A., Nonaka, D., Takeuchi, R., de Los Reyes, C., Uehara, M., Khampheng, P., et al. (2023). Trend of sociodemographic and economic inequalities in the use of maternal health services in Lao People's Democratic Republic from 2006 to 2017: MICS data analysis. *Tropical Medicine and Health, 51*(1), 56. doi:10.1186/s41182-023-00548-2

Akkhavong, K., Paphassarang, C., Phoxay, C., Vonglokham, M., Phommavong, C., and Pholsena, S. (2014). *Lao People's Democratic Republic Health System Review*. Manila: Asia Pacific Observatory on Health Systems.

Alkenbrack, S., Hanson, K., and Lindelow, M. (2015). Evasion of "mandatory" social health insurance for the formal sector: evidence from Lao PDR. *BMC Health Services Research, 15*, 473. doi:10.1186/s12913-015-1132-5

Alkenbrack, S., Jacobs, B., and Lindelow, M. (2013). Achieving universal health coverage through voluntary insurance: what can we learn from the experience of Lao PDR? *BMC Health Services Research, 13*(1), 521. doi:10.1186/1472-6963-13-521

Alkenbrack, S., and Lindelow, M. (2013). The Impact of Community-Based Health Insurance on Utilization and Out-of-Pocket Expenditures in Lao People's Democratic Republic. *Health Economics, 24*(4), 379–399. doi:doi/10.1002/hec.3023

Arellano, M. (1987). Computing robust standard errors for within-groups estimators. *Oxford Bulletin of Economics and Statistics, 49*(4).

Arsenault, C., Gage, A., Kim, M. K., Kapoor, N. R., Akweongo, P., Amponsah, F., et al. (2022). COVID-19 and resilience of healthcare systems in ten countries. *Nature Medicine, 28*(6), 1314–1324. doi:10.1038/s41591-022-01750-1

Aryal, A., Clarke-Deelder, E., Phommalangsy, S., Kounnavong, S., and Fink, G. (2024a). Health system inequities in Lao People's Democratic Republic: Evidence from a nationally representative phone survey. *Tropical Medicine & International Health, 29*(6), 518–525. doi:10.1111/tmi.13997

Aryal, A., Clarke-Deelder, E., Phommalangsy, S., Kounnavong, S., and Fink, G. (2024b). Health system utilization and perceived quality among adults in Lao PDR: evidence from a nationally representative phone survey. *BMC Public Health, 24*(1), 565. doi:10.1186/s12889-024-18039-2

Bodhisane, S., and Pongpanich, S. (2015). The Impact of Community Based Health Insurance in Enhancing Better Accessibility and Lowering the Chance of Having Financial Catastrophe Due to Health Service Utilization: A Case Study of Savannakhet Province, Laos. *International Journal of Health Services, 47*(3), 504–518. doi:10.1177/0020731415595609

Bodhisane, S., and Pongpanich, S. (2017). Community based health insurance: A case study. *British Journal of Health Care Management, 23*(2), 74–82. doi:10.12968/bjhc.2017.23.2.74

Bodhisane, S., and Pongpanich, S. (2019a). Factors affecting the willingness to join community-based health insurance (CBHI) scheme: A case study survey from Savannakhet Province, Lao P.D.R. *International Journal of Health Policy and Management, 34*(2), 604–618.

Bodhisane, S., and Pongpanich, S. (2019b). The impact of National Health Insurance upon accessibility of health services and financial protection from catastrophic health expenditure: a case study of Savannakhet province, the Lao People's Democratic Republic. *Health Research Policy and Systems, 17*(1), 99. doi:10.1186/s12961-019-0493-3

Bodhisane, S., and Pongpanich, S. (2022). The influence of the National Health Insurance scheme of the Lao People's Democratic Republic on healthcare access and catastrophic health expenditures for patients with chronic renal disease, and the possibility of integrating organ transplantation into the health financing system. *Health Research Policy and Systems, 20*(1), 71. doi:10.1186/s12961-022-00869-4

Bodhisane, S., and Pongpanich, S. (2023). Laos' Social Health Insurance (SHI) program's impact on older people's accessibility and financial security against catastrophic health expense. *BMC Health Services Research, 23*(1), 1317. doi:10.1186/s12913-023-10063-z

Bottomley, C., Scott, J. A. G., and Isham, V. (2019). Analysing interrupted time series with a control. *Epidemiologic Methods, 8*(1), 20180010. doi:10.1515/em-2018-0010

Boudreaux, C., Chanthala, P., and Lindelow, M. (2014). Assessing the elimination of user fees for delivery services in Laos. *PloS one, 9*(3), e89784. doi:10.1371/journal.pone.0089784

Canadian Agency for Drugs and Technologies in Health. RCT / CCT - PubMed. *CADTH Search Filters Database* Retrieved from https://searchfilters.cadth.ca/link/108

Canadian Agency for Drugs and Technologies in Health. SR / MA / HTA / ITC - PubMed. *CADTH Search Filters Database* Retrieved from https://searchfilters.cadth.ca/link/99

Chaleunvong, K., Phoummalaysith, B., Phonvixay, B., Sychareun, V., Durham, J., and Essink, D. (2020). Factors affecting knowledge of National Health Insurance Policy among out-patients in Lao PDR: an exit interview study. *Global Health Action, 13*(sup2), 1791414. doi:10.1080/16549716.2020.1791414

Chaleunvong, K., Phoummalaysith, B., Phonvixay, B., Vonglokham, M., Sychareun, V., Durham, J., et al. (2020). Factors associated with patient payments exceeding National Health Insurance fees and out-of-pocket payments in Lao PDR. *Global Health Action, 13*(sup2), 1791411. doi:10.1080/16549716.2020.1791411

Chu, A., Phommavong, C., Lewis, J., Braa, J., and Senyoni, W. (2017). *Applying ICT to health information systems (HIS) in low resource settings: implementing DHIS2 as an integrated health information platform in Lao PDR.* Paper presented at the International Conference on Social Implications of Computers in Developing Countries.

Cochrane Effective Practice and Organisation of Care. *EPOC Low- and middle-income country (LMIC) filters – December 2022*.

Coulombe, H., Epprecht, M., Pimhidzai, O., and Sisoulath, V. (2016). *Where are the poor? Lao PDR 2015 census-based poverty map: province and district level results*. Vientiane: Lao Statistics Bureau.

Erlangga, D., Suhrcke, M., Ali, S., and Bloor, K. (2019). The impact of public health insurance on health care utilisation, financial protection and health status in low-and middle-income countries: A systematic review. *PloS one, 14*(8), e0219731. doi:10.1371/journal.pone.0219731

Friedman, J. H. (1984). *A variable span smoother*: Laboratory for Computational Statistics, Department of Statistics, Stanford University Technical Report No. 5.

Prime Ministrial Decree on Medical Services (No.52), (1995).

International Labour Organization. (2021). *Extending Social Health Protection in Lao People's Democratic Republic: Accelerating progress towards Universal Health Coverage*. Bangkok: International Labour Organization.

International Labour Organization. (2024). *Understanding informality and expanding social security coverage in the Lao People's Democratic Republic*. Vientiane: International Labour Organization.

Ito, T., Kounnavong, S., and Miyoshi, C. (2022). Financial burden and health-seeking behaviors related to chronic diseases under the National Health Insurance Scheme in Bolikhamxay Province, Lao PDR: a cross-sectional study. *International Journal for Equity in Health, 21*(1), 180. doi:10.1186/s12939-022-01788-0

Jeffery, T., Ryotaro, H., and Yunsun, L. (2015). *Improving Coverage and Utilization of Maternal and Child Health Services in Lao PDR: Impact Evaluation of the Community Nutrition Project*. Washington, DC: World Bank.

Kaiser, A. H., Rotigliano, N., Flessa, S., Ekman, B., and Sundewall, J. (2023). Extending universal health coverage to informal workers: A systematic review of health financing schemes in low- and middle-income countries in Southeast Asia. *PloS one, 18*(7), e0288269. doi:10.1371/journal.pone.0288269

Kontopantelis, E., Doran, T., Springate, D. A., Buchan, I., and Reeves, D. (2015). Regression based quasi-experimental approach when randomisation is not an option: interrupted time series analysis. *BMJ, 350*, h2750. doi:10.1136/bmj.h2750

Lao Health Costing Commitee. (2014). *Costing of public health care facilities in Lao PDR*. Vientiane: ILO-WHO Project on 'Supporting the Establishment of the National Health Insurance Scheme and the Extension of Coverage.

Lao Statistics Bureau. (2015). *The 4th Population and Housing Census (PHC) 2015 Lao PDR*. Vientiane: Lao Statistics Bureau.

Lopez Bernal, J., Cummins, S., and Gasparrini, A. (2018). The use of controls in interrupted time series studies of public health interventions. *International Journal of Epidemiology, 47*(6), 2082–2093. doi:10.1093/ije/dyy135

MEASURE Evaluation. (2020). *Considerations for the Use of Routine Data for Evaluation of Public Health Programs*.

National Guideline on Revolving Drug Funds in Public Health Facilities, (1997).
Ministry of Health Lao PDR. (2016). *National Health Statistics Report FY 2014-2015*. Vientiane: Ministry of Health Lao PDR.
Instruction on the collection of contributions, payment mechanisms and calculation of service fees of the National Health Insurance scheme (No. 0476), (2018).
Ministry of Health Lao PDR. (2021). *National Health Accounts Report Fiscal Year 2019*. Vientiane: Ministry of Health Lao PDR.
Ministry of Health Lao PDR. (2022). *National Health Statistics Report 2020*. Vientiane: Ministry of Health Lao PDR.
Murakami, H., Phommasack, B., Oula, R., and Sinxomphou, S. (2001). Revolving drug funds at front-line health facilities in Vientiane, Lao PDR. *Health policy and planning, 16*(1), 98–106. doi:10.1093/heapol/16.1.98
Nagpal, S., Masaki, E., Pambudi, E. S., and Jacobs, B. (2019). Financial protection and equity of access to health services with the free maternal and child health initiative in Lao PDR. *Health policy and planning, 34*(Supplement_1), i14–i25. doi:10.1093/heapol/czz077
National Institute of Public Health Lao PDR. (2005). *Report on the formulation process on user fees and exemption policy in the context of Lao PDR (1975-2005)*. Vientiane: National Institute of Public Health Lao PDR.
National Institute of Public Health Lao PDR. (2017). *Analysis of financial burden of out-of-pocket patient payments in Lao PDR using LECS 5 data*. Vientiane: National Institute of Public Health Lao PDR.
Phommasack, B., Oula, L., Khounthalivong, O., Keobounphanh, I., Misavadh, T., Loun, et al. (2005). Decentralization and recentralization: effects on the health systems in Lao PDR. *The Southeast Asian Journal of Tropical Medicine and Public Health, 36*(2), 523–528.
Rambachan, A., and Roth, J. (2023). A More Credible Approach to Parallel Trends. *Review of Economic Studies, 90*(5), 2555–2591. doi:10.1093/restud/rdad018
Rannan-Eliya, R. P., Anuranga, C., Chandrasiri, J., Hafez, R., Kasthuri, G., Wickramasinghe, R., et al. (2012). *Impact of out-of-pocket expenditures on families and barriers to use of maternal and child health services in Asia and the Pacific: Evidence from national household surveys of healthcare use and expenditures—summary technical report*. Manila: Asian Development Bank.
Sydavong, T., Goto, D., Kawata, K., Kaneko, S., and Ichihashi, M. (2019). Potential demand for voluntary community-based health insurance improvement in rural Lao People's Democratic Republic: A randomized conjoint experiment. *PloS one, 14*(1), e0210355. doi:10.1371/journal.pone.0210355
Thomé, J.-M., and Pholsena, S. (2009). Lao People's Democratic Republic: Health Financing Reform and Challenges in Expanding the Current Social Protection Schemes. In *Promoting sustainable strategies to improve access to health care in the Asian and Pacific Region* (pp. 1–32). Bangkok: United Nation ESCAP.
Tukey, J. W. (1977). *Exploratory data analysis* (Vol. 2). Reading, MA: Addison-Wesley.
Turner, H. C., Lauer, J. A., Tran, B. X., Teerawattananon, Y., and Jit, M. (2019). Adjusting for Inflation and Currency Changes Within Health Economic Studies. *Value in Health, 22*(9), 1026–1032. doi:10.1016/j.jval.2019.03.021
Wagenaar, B. H., Sherr, K., Fernandes, Q., and Wagenaar, A. C. (2016). Using routine health information systems for well-designed health evaluations in low- and middle-income countries. *Health policy and planning, 31*(1), 129–135. doi:10.1093/heapol/czv029
World Bank. (2010a). *Community-based health insurance in Lao PDR: understanding enrollment and impact*. Washington, DC: World Bank.
World Bank. (2010b). *Enrollment of firms in social security in Lao PDR: perspectives from the private sector*. Washington, DC: World Bank.
World Bank. (2010c). *Out-of-pocket spending and health service utilization in Lao PDR: evidence from the Lao Expenditure and Consumption Surveys*. Washington, DC: World Bank.
World Bank. (2013). *Maternal and Child Health Out-of-Pocket Expenditure and Service Readiness in Lao PDR*. Washington, DC: World Bank.
World Bank. (2015). *Drivers of poverty reduction in Lao PDR* (Vol. World Bank): Washington, DC.
World Bank. (2017a). *Lao Poverty Policy Brief: Taxonomy of the poor and its usefulness in policy design*. Washington, DC: World Bank.

World Bank. (2017b). *Managing Transition: Reaching the Vulnerable while Pursuing Universal Health Coverage*. Washington, DC: World Bank.
World Bank. (2021). World Bank Country and Lending Groups.
World Health Organization Regional Office for the Western Pacific. (2023). *Progress towards Universal Health Coverage in the Lao People's Democratic Republic: Monitoring financial protection 2007–2019*. Manila: World Health Organization Regional Office for the Western Pacific.